Variational Quantum Eigensolver for a CO_2 Molecule

Team Name

- Alpha Squadron

Members

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Submission Category

- Rising Star

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Summary

- Researched the applications of VQE beyond the lecture.
- Decided on the topic of using VQE to simulate a CO2 molecule.
- Researched Qiskit packages.
- Created code using VQE to calculate the total ground state energy of CO2.
- The total energy at different bond lengths were compared to find the equilibrium bond length.

Motivation

- Our group was interested in VQE's application in chemistry.
- During our research into VQE we decided to focus on the application of solving for the ground state of a triatomic molecule. We chose CO2 because of its linear geometry.

Code Workflow

- PySCF driver ➡ Active Space Reduction ➡ Jordan-Wigner & Parity Mapping ➡ UCCSD Ansatz ➡ VQE solver
- Reduced the number of qubits by applying Active Space Transformation, Jordan-Wigner Mapping, and Parity Mapping.

Results

- The total ground state energy for CO_2 was calculated using VQE in the 6-31G basis set as -187.559 Ha compared to the accepted value of -187.651 Ha. The error for the 6-31G basis set was 0.0297% against the value reported by Hassan and Abood while the error when comparing the energy calculated in the STO-3G basis set was 0.0394%. Therefore, the 6-31G basis was used to calculate the total ground state energies in the VQE program at different bond lengths, shown in Table 1.
- The total ground-state energy comprises the electronic and nuclear contributions. At very large distances, the C and O atoms decouple from each other, and the energy approaches the sum of the separated atoms' energies. At short distances, the nuclear repulsion diverges. We find an equilibrium bond length of 1.16 Å, where electron-nuclear attraction balances the nucleus-nucleus repulsion.

Table 1. Total ground state energy in Hartrees for CO_2 at bond lengths in Angstroms calculated using VQE in the 6-31G basis set.

Bond Length (Å)	**Total Ground State Energy (Ha)**
0.58	–180.370
0.62	–182.147
0.67	–183.600
0.72	–184.761
0.77	–185.664
0.82	–186.345
0.88	–186.839
0.94	–187.178
1.01	–187.388
1.084	–187.527
1.16	**–187.559**
1.25	–187.534
1.33	–187.470
1.43	–187.382
1.53	–187.286
1.64	–187.194
1.76	–187.115
1.89	–187.015
2.02	–187.015
2.19	–186.987
2.32	–186.967

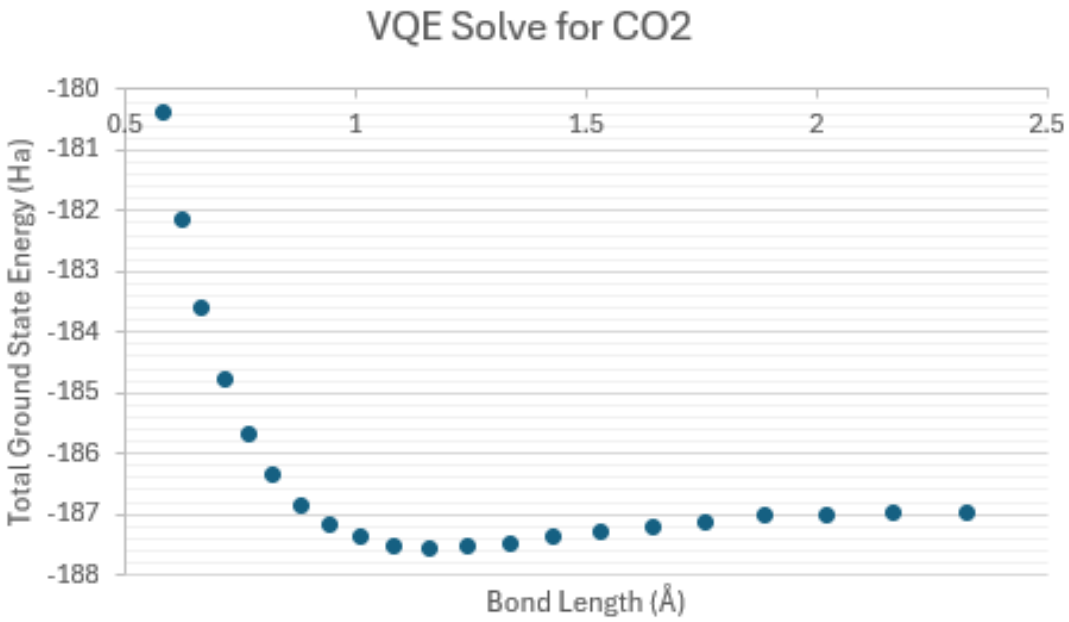


Figure 1. VQE total ground state energies (Ha) as a function of bond length (Å) for CO_2. The equilibrium bond distance was calculated at 1.16 Å with a minimum total ground state energy of –187.559 Ha.

References

- Ground state solvers - Qiskit Nature 0.8.0. Github.io2018. Accessed October 24, 2026. Ground state solvers - Qiskit Nature 0.8.0

- Hasan E, Abood S. Utilizing the Hartree-Fock Method to Analyze Carbon Dioxide Molecule. *American Journal of Physical Chemistry*2024;13(2):1281211. doi:10.11648/j.ajpc.20241302.13

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